

# Denis Vinokurov

Ph.D. Candidate in Mathematics

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## Education

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|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| 2027 (expected) | Ph.D. in Mathematics<br>• Advisors: I. Polterovich, M. Karpukhin                                                                                      | <i>Université de Montréal</i>           |
| 2024            | M.Sc. in Mathematics<br>• Diploma with honors<br>• Thesis: “The first eigenvalue of the Laplacian on orientable surfaces”<br>• Advisor: A. V. Penskoi | <i>Independent University of Moscow</i> |
| 2022            | B.Sc. in Mathematics<br>• Diploma with honors<br>• Thesis: “Geometry of harmonic maps of superminimal tori into spheres”<br>• Advisor: A. V. Penskoi  | <i>HSE University</i>                   |

## Research Interests

Geometric analysis and differential geometry, with particular interests in spectral geometry, eigenvalue optimization, harmonic maps, and minimal submanifolds; variational and nonsmooth analysis.

## Honors and Awards

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|----------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------|
| 2026 – Present | Canada Graduate Research Scholarship – Doctoral (CGRS D)              | <i>Natural Sciences and Engineering Research Council of Canada (NSERC)</i> |
| 2026           | Doctoral Research Scholarship<br>awarded; declined in favor of CGRS D | <i>Fonds de recherche du Québec (FRQNT)</i>                                |
| 2023 – 2025    | ISM Graduate Scholarship                                              | <i>Institut des Sciences Mathématiques (ISM)</i>                           |

## Publications and Preprints

- D. Vinokurov. “Eigenvalue optimization via a first-variation formula”. 2026. arXiv: [2606.31869 \[math.SP\]](#). Submitted
- D. Vinokurov. “Geometric bounds for Steklov and weighted Neumann eigenvalues on Euclidean domains”. 2026. arXiv: [2604.03418 \[math.SP\]](#). Submitted, revisions requested
- D. Vinokurov. “Maximizing higher eigenvalues in dimensions three and above”. 2025. arXiv: [2506.09328 \[math.SP\]](#). Submitted
- D. Vinokurov. “Eigenvalue optimization in higher dimensions and  $p$ -harmonic maps”. **Geom. Funct. Anal.** 36.4 (2026), pp. 1244–1293. DOI: [10.1007/s00039-026-00751-3](#)
- D. Vinokurov. “Conformal optimization of eigenvalues on surfaces with symmetries”. **J. Lond. Math. Soc.** 112.6 (2025), e70386. DOI: [10.1112/jlms.70386](#)
- M. Karpukhin and D. Vinokurov. “The first eigenvalue of the Laplacian on orientable surfaces”. **Math. Z.** 301.3 (2022), pp. 2733–2746. DOI: [10.1007/s00209-022-03009-4](#)

## Conference Talks

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|-----------------|------------------------------------------------------------------|-------------------------------|
| <i>Feb 2026</i> | Workshop: <i>Emerging Horizons in Geometric Spectral Theory</i>  | <i>Isaac Newton Institute</i> |
| <i>Dec 2025</i> | <i>Winter Meeting of the Canadian Mathematical Society</i>       | <i>Toronto</i>                |
| <i>Jun 2025</i> | Workshop: <i>Geometric Problems in Linear and Nonlinear PDEs</i> | <i>Isle of Skye, Scotland</i> |
| <i>Dec 2023</i> | <i>Winter Meeting of the Canadian Mathematical Society</i>       | <i>Montréal</i>               |

## Selected Seminar Talks

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| <i>Feb 2026</i> | <i>Two informal talks on Palais–Smale sequences in eigenvalue optimization</i> | <i>Isaac Newton Institute</i>                    |
| <i>Feb 2026</i> | <i>Functional Analysis and Noncommutative Geometry Seminar</i>                 | <i>HSE University (online)</i>                   |
| <i>Nov 2025</i> | <i>Spectral Geometry in the Clouds</i>                                         | <i>Online</i>                                    |
| <i>Oct 2025</i> | <i>CRM/Montreal/Quebec Analysis Seminar</i>                                    | <i>Université de Montréal</i>                    |
| <i>Oct 2025</i> | <i>Analysis Seminar</i>                                                        | <i>Cornell University (online)</i>               |
| <i>Oct 2025</i> | <i>Spectral Geometry Seminar</i>                                               | <i>Independent University of Moscow (online)</i> |
| <i>Feb 2025</i> | <i>CIRGET Junior Seminar</i>                                                   | <i>UQAM</i>                                      |

## Participation in Schools and Workshops

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| <i>May 2026</i> | <i>Conference on Classical Analysis in Memory of Paul Koosis</i>                        | <i>Université de Montréal</i> |
| <i>Jul 2025</i> | ISM Discovery School: <i>Interactions between Convex Geometry and Spectral Analysis</i> | <i>Université de Montréal</i> |
| <i>Jun 2023</i> | ISM Discovery School: <i>Geometry and Spectra of Random Hyperbolic Surfaces</i>         | <i>Université de Montréal</i> |
| <i>May 2023</i> | Workshop: <i>Spectral Geometry and Applications</i>                                     | <i>Université Laval</i>       |

## Teaching Experience

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|--------------------|-------------------------------------------|-----------------------------------------|
| <i>Fall 2025</i>   | Teaching Assistant, Linear Algebra        | <i>Université de Montréal</i>           |
| <i>Winter 2025</i> | Teaching Assistant, Euclidean Geometry    | <i>Université de Montréal</i>           |
| <i>Fall 2024</i>   | Teaching Assistant, Differential Geometry | <i>Université de Montréal</i>           |
| <i>Spring 2022</i> | Teaching Assistant, Differential Geometry | <i>Independent University of Moscow</i> |
| <i>Fall 2021</i>   | Teaching Assistant, Analysis on Manifolds | <i>HSE University</i>                   |
| <i>Fall 2020</i>   | Teaching Assistant, Analysis on Manifolds | <i>HSE University</i>                   |

## Mentorship

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| <i>Summer 2026</i> | Joint supervision of a CRM–ISM undergraduate research project with Prof. Galia Dafni | <i>Université de Montréal</i> |
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## Refereeing

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Geometric and Functional Analysis; Journal of the London Mathematical Society; Mathematische Zeitschrift.

## Languages

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Russian (native), English (fluent), French (intermediate).